



Technical Progress Report

Project name:	Biped robot		
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1. Executive Summary

This report summarizes the technical progress of the biped robot project this week. The team found that the original stepping motion was too loose, leading to unstable leg movement, so the stepping gait has been redesigned for better stability. In addition, the current hardware compartment exposes too many components, which brings potential damage risks if the robot falls. Therefore, the hardware compartment needs to be redesigned. For vision recognition, the camera has difficulty distinguishing cubes and square pyramids at specific angles. The team is collecting more training data to improve recognition accuracy.

2. Objectives & Scope

- Redesign the stepping gait to solve the loose and unstable leg movement during motion.
- Improve the structural protection of the hardware compartment and reduce exposed components.
- Analyze recognition errors between cubes and square pyramids at specific viewing angles.
- Collect more image datasets to train the vision model for better accuracy.
- Prepare for more stable walking tests after structural and gait optimization.

3. Technical Work Completed

3.1 Gait Optimization

Observed unstable leg movement caused by overly loose original stepping motion. Redesigned the stepping trajectory and joint motion sequence for higher rigidity and stability.

3.2 Hardware Compartment Inspection

Checked the current hardware compartment layout. Confirmed that excessive exposed components (ESP32, camera, battery, wires) pose risks of damage from falling or collision.

3.3 Vision Recognition Testing

Tested shape recognition performance at different angles. Found that the camera cannot reliably distinguish cubes and square pyramids at specific viewpoints.

3.4 Dataset Preparation

Started collecting more images of cubes and square pyramids from multiple angles to expand the training dataset.

4. Observations & Findings

- The original stepping design is too loose, resulting in weak support and unstable leg movement during walking attempts.
- Excessive components are exposed outside the hardware compartment, which may be damaged if the robot falls during movement.
- The vision system shows confusion between cubes and square pyramids at specific angles.
- Insufficient training data leads to low robustness of the shape recognition model.

5. Challenges & Roadblocks

- Unstable and loose stepping motion prevents reliable walking tests.
- Exposed components lack mechanical protection, limiting further motion experiments.
- Similar appearance of cubes and square pyramids causes recognition errors at certain angles.
- Limited training data leads to poor angle adaptability of the vision model.

6. Next Steps & Recommendations

- Finalize and verify the redesigned stepping gait to ensure stable and firm leg movement.
- Redesign the hardware compartment to fully enclose and protect electronic components.
- Continue collecting more multi-angle image data for model training.
- Retrain the vision recognition model to improve accuracy between cubes and square pyramids.
- Integrate the optimized structure and gait, then conduct stable walking tests.

7. Conclusion

This week, the team identified key issues in gait, structure and vision recognition. The original stepping motion is too loose for stable movement, so a new stepping design has been developed. The exposed hardware compartment requires redesign to avoid component damage. Meanwhile, the vision system has difficulty distinguishing cubes and square pyramids at specific angles, so more data will be collected for retraining. Next week, the team will focus on gait verification, hardware compartment redesign, and dataset expansion to improve the robot's motion stability and recognition robustness.